Highlights

Hidden Behind the Mean: Probabilistic Thermal-Sensation Diagnostics of Discomfort-Tail Risk Under Future Weather

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- Audits discomfort-tail probability in 144 fixed-archetype future-weather cases.
- Tests where scalar and spatial summaries lose tail-risk information.
- Finds mean-zone high-tail exceedance of 13.4% versus 37.0% under any-zone screening.
- Shows the zone-aggregation gap persists across p_{tail} thresholds and no-PMV robustness checks.
- Finds metabolic-rate assumptions change max-zone high-tail status in 38.7% of occupied records.

Hidden Behind the Mean: Probabilistic Thermal-Sensation Diagnostics of Discomfort-Tail Risk Under Future Weather

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Abstract

Scalar comfort summaries such as PMV or expected thermal sensation can hide probability mass assigned to severe cold or warm sensation categories. This study presents a diagnostic future-weather stress test for a fixed DOE Medium Office archetype operated with a fixed reference schedule. Thermal sensation vote (TSV) is represented as a calibrated seven-class probability distribution, and discomfort-tail probability is defined as $p_{\mathrm{tail}} = \Pr(\mathrm{TSV} \leq -2) + \Pr(\mathrm{TSV} \geq +2)$. This quantity is used as a diagnostic screen, not as a validated action rule or universal threshold. The building was simulated under 144 weather files spanning six cities, two emissions scenarios, four time slices, and three annual severities. The occupant model and building prototype are held fixed, so the experiment is a conditional stress test rather than a forecast of future occupants or buildings. Across 2.41 million occupied timesteps, mean-zone high-tail exceedance at $p_{\mathrm{tail}} \geq 0.20$ was 13.4%, with a 95th-percentile p_{tail} of 0.306. Expected TSV tracked p_{tail} closely, while PMV was not interchangeable with the tail-probability screen. Zone aggregation produced the largest information loss: mean aggregation gave 13.4% modeled high-tail exceedance, while any-zone aggregation gave 37.0%, hiding at least one high-tail zone in 23.6% of occupied timesteps. This gap persisted across threshold checks. A metabolic-spread diagnostic showed that representative metabolic assumptions changed max-zone high-tail status in 38.7% of occupied timesteps. Calibrated TSV probability states therefore provide an auditable diagnostic interface for tail-risk information that scalar means, representative occupants, and spatial averages can obscure.

Keywords: probabilistic thermal comfort, thermal sensation vote, discomfort risk, PMV, future weather, EnergyPlus, occupant-centric diagnostics

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1. Introduction

Building comfort diagnostics depend on how occupant state is represented. A scalar value such as Predicted Mean Vote (PMV), expected thermal sensation, or a binary acceptability flag is easy to communicate and easy to compare across cases. It is also lossy. A scalar mean can indicate the center of a predicted comfort state while hiding how much probability lies in the discomfort tails of the thermal sensation scale.

This distinction matters because thermal sensation is not deterministic. The same air temperature, radiant temperature, humidity, clothing level, and metabolic assumption can correspond to different thermal sensation votes (TSVs) across people and contexts. Field data show substantial variation around any single comfort index, and that variation is not only noise; it reflects differences in adaptation, expectation, personal physiology, recent exposure, and behavior [1, 2, 3, 4]. A diagnostic that records only the mean predicted state cannot directly show whether the predictor assigns nontrivial probability to severe cold or warm sensation categories.

This paper therefore treats a predicted TSV distribution as an information object. For each occupied timestep, the predictor returns probabilities for the seven ASHRAE sensation classes $k \in \{-3, -2, -1, 0, +1, +2, +3\}$. The expected sensation,

$$\mu_{\mathrm{TSV}} = \sum_{k=-3}^{3} k p_k, \tag{1}$$

summarizes the central tendency of that distribution. The discomfort-tail probability,

$$p_{\mathrm{tail}} = \Pr(\mathrm{TSV} \leq -2) + \Pr(\mathrm{TSV} \geq +2), \tag{2}$$

summarizes the probability mass assigned to the cold and warm discomfort tails. In plain terms, p_{tail} is the model-estimated probability that an occupant state falls in the "cool/cold" or "warm/hot" sides of the TSV scale, rather than in the neutral or mildly cool/warm bins. It is not PMV, not PPD, and not a measured dissatisfaction fraction. It is a diagnostic summary of a calibrated categorical prediction.

The analysis uses future weather as a stress-test domain. Hotter and more variable weather can push mechanically conditioned buildings toward higher warm-tail probability, but this study does not claim to forecast how occupants in 2050 or 2080 will feel. The occupant model is held fixed. The DOE Medium Office prototype and reference thermostat schedule are also held fixed. These choices make the experiment a controlled conditional stress test: if the same current archetype and the same comfort-risk mapping are evaluated under selected warming-climate weather files, what risk information becomes visible in the probability distribution that scalar summaries and spatial averages can miss?

This framing is deliberately diagnostic. The paper does not validate HVAC operation, optimize setpoints, claim energy savings, or test actuator feasibility. Building operation is a downstream motivation: an automation system can use only the information made available to it. The present question is narrower: whether a calibrated TSV probability state exposes discomfort-tail risk that scalar comfort states, representative occupant assumptions, and mean-zone summaries hide.

The contributions are:

- a clear diagnostic definition of p_{tail} from a calibrated seven-class TSV probability distribution;
- a 144-case fixed-archetype Medium Office future-weather stress test across six cities, two emissions scenarios, four time slices, and three annual severities;
- scalar-state audits comparing p_{tail} with expected TSV and mean PMV;
- a zone-aggregation audit showing how mean-zone probability summaries hide modeled any-zone high-tail exceedance;
- a metabolic-spread diagnostic showing how representative metabolic assumptions change estimated tail-risk status.

The broader claim is narrow: probabilistic TSV distributions are useful because they make occupant-state uncertainty inspectable. They do not remove the need for adaptive-comfort theory, better occupant models, building-specific validation, or future operational studies.

2. Background

2.1. Scalar comfort indices and what they compress

PMV remains the most widely recognized steady-state comfort index for mechanically conditioned buildings [5, 6, 7]. Adaptive comfort models provide another scalar framework by relating acceptable indoor temperatures to recent outdoor conditions [1, 2, 3]. These models are valuable for standards, simulation diagnostics, and building-performance comparison. Their strength is also their limitation: they compress a distributed occupant response into a scalar value, band, or violation state.

That compression is acceptable when the question is whether an average condition is near a standard reference. It is less complete when the question is whether severe sensation is probable. A scalar value cannot show whether probability mass is concentrated near neutral, spread across mild categories, or split between warm and cold discomfort tails. This paper does not argue that PMV or expected TSV should be discarded. It asks what additional information becomes visible when the full TSV probability distribution is retained.

2.2. Probabilistic and ordinal TSV modeling

The ASHRAE Global Thermal Comfort Database II provides a large field dataset linking environmental and personal variables to observed thermal sensation votes [4]. Because TSV is ordered categorical rather than continuous, ordinal probabilistic models are a natural fit [8, 9, 10]. They can preserve the order of the scale while producing class probabilities for each sensation category. Probability calibration is then needed because raw classifier scores are not automatically reliable probabilities [11, 12].

Recent comfort-modeling studies increasingly use probabilistic or Bayesian representations to express uncertainty in occupant response [13, 14, 15]. The contribution here is not a new claim that probabilistic comfort modeling exists. The contribution is a diagnostic use of the distribution: expected TSV and discomfort-tail probability are compared directly under future-weather stress, and the information loss from scalar and spatial aggregation is quantified.

2.3. Future-weather stress tests and fixed mappings

Climate-informed building simulation often uses generated or selected future weather files to test how current building assumptions respond under altered outdoor forcing [16, 17]. Overheating and climate-responsive envelope studies similarly use future scenarios to expose vulnerability, while recognizing that actual future risk will depend on adaptation, envelope changes, operating practices, and occupant behavior [18, 19].

This study follows that stress-test logic. It does not assume that 2080 occupants have the same expectations, clothing behavior, physiology, or adaptive opportunities as current field observations. It holds the comfort-risk mapping fixed so the diagnostic question remains controlled. The result is conditional: under selected future-weather files, with a fixed prototype and fixed occupant model, how often do scalar summaries and mean aggregation hide modeled discomfort-tail exceedance?

2.4. From diagnostic states to downstream uses

Probabilistic comfort states could be used by many downstream decision frameworks, including rule-based supervisory policies, stochastic MPC, fuzzy rule bases, or operator dashboards. This paper does not compare those frameworks and does not claim stability, optimality, or actuator feasibility. Its scope ends at the diagnostic interface: the probability distribution, the scalar and tail summaries derived from it, and the audits that show where information is lost.

3. Methods

3.1. Thermal-comfort data and TSV probability model

The TSV probability model follows the published ordinal thermal-comfort framework of Guo and Aviv [20] and was trained from the ASHRAE Global Thermal Comfort Database II [4]. Records with incomplete core fields were removed, environmental variables were screened for extreme outliers, and the remaining observations were split into training, calibration, and held-out test partitions using stratified sampling over the seven TSV classes. The feature set included air temperature, mean radiant temperature, relative humidity, air speed, clothing insulation, metabolic rate, body surface area, PMV, and adaptive-comfort descriptors. PMV is included as a covariate because it is a useful physics-based summary of the same thermal state; the supervised target remains the observed TSV.

The deployed model is an ordinal gradient-boosted predictor with post-hoc isotonic calibration of cumulative probabilities. A nominal multiclass model was retained as a sensitivity comparator during model development, but the main diagnostic uses the ordinal predictor because it respects the ordered TSV scale. Calibration and label-support checks are reported in the supplementary material. Extreme TSV labels are sparser than neutral and mild labels, so tail probabilities should be interpreted as calibrated diagnostic estimates rather than as direct measurements of occupant dissatisfaction.

3.2. Probability object and definition of p_{tail}

For each occupied timestep t, the predictor returns a probability vector over the seven TSV classes:

$$\mathbf{p}^{(t)} = \{p_{-3}^{(t)}, p_{-2}^{(t)}, p_{-1}^{(t)}, p_{0}^{(t)}, p_{+1}^{(t)}, p_{+2}^{(t)}, p_{+3}^{(t)}\}, \tag{3}$$

where $p_k^{(t)} = \Pr(\text{TSV} = k \mid \mathbf{x}^{(t)})$ and $\sum_k p_k^{(t)} = 1$.

The expected sensation is

$$\mu_{\text{TSV}}^{(t)} = \sum_{k=-3}^{+3} k p_k^{(t)}. \tag{4}$$

It is a scalar mean of the predicted categorical distribution. Positive values indicate warmer expected sensation; negative values indicate cooler expected sensation.

The discomfort-tail probability is

$$p_{\text{tail}}^{(t)} = \Pr(\text{TSV} \leq -2 \mid \mathbf{x}^{(t)}) + \Pr(\text{TSV} \geq +2 \mid \mathbf{x}^{(t)}) = p_{-3}^{(t)} + p_{-2}^{(t)} + p_{+2}^{(t)} + p_{+3}^{(t)}. \tag{5}$$

This is the central diagnostic variable in the paper. It is the probability mass assigned to the two discomfort tails of the TSV scale. The lower tail contains cool and cold votes; the upper tail contains warm and hot votes. Neutral and mild votes $(-1, 0, +1)$ are not counted in p_{tail}.

We also record warm-tail probability,

$$p_{\text{warm}}^{(t)} = p_{+2}^{(t)} + p_{+3}^{(t)}, \tag{6}$$

cold-tail probability,

$$p_{\text{cold}}^{(t)} = p_{-3}^{(t)} + p_{-2}^{(t)}, \tag{7}$$

and directional tail dominance,

$$d_{\text{tail}}^{(t)} = p_{\text{warm}}^{(t)} - p_{\text{cold}}^{(t)}. \tag{8}$$

The screening threshold $p_{\text{tail}} \geq 0.20$ is used to label a timestep as high-tail in the diagnostic tables. It is not a universal comfort-preference threshold, not an ASHRAE acceptability rule, and not an operational action threshold. It loosely corresponds to a state where at least 20% of the predicted probability mass lies in the discomfort tails, leaving at most 80% in neutral or mild categories. Because the numerical exceedance share depends on this screen, we also audit the main aggregation result over thresholds from 0.05 to 0.40.

Table 1: Diagnostic quantities derived from the TSV probability distribution.

Quantity	**Definition**	**Role in this study**
p_k	$\Pr(\text{TSV} = k)$, $k \in \{-3, \ldots, +3\}$	Full categorical comfort-state distribution.
μ_{TSV}	$\sum_k kp_k$	Scalar expected sensation used as a mean-state comparator.
p_{tail}	$p_{-3} + p_{-2} + p_{+2} + p_{+3}$	Total probability assigned to severe cold or warm sensation bins.
p_{warm}	$p_{+2} + p_{+3}$	Warm-side component of the tail.
p_{cold}	$p_{-3} + p_{-2}$	Cold-side component of the tail.
d_{tail}	$p_{\text{warm}} - p_{\text{cold}}$	Directional tail dominance; positive values indicate warm-tail dominance.

3.3. Fixed building and future-weather diagnostic panel

The simulation testbed is the DOE Medium Office prototype [21] simulated in EnergyPlus v25.1 [22]. The building envelope, HVAC system, schedules, and setpoint logic are fixed across all weather cases. During occupied weekdays from 06:00 to 22:00, the reference schedule maintains fixed occupied heating and cooling setpoints of 22 and 24 °C. During unoccupied hours, setpoints are relaxed. The probability model is evaluated every 15 minutes from simulated zone states, but no probability-derived setpoint action is applied in this diagnostic run. The panel therefore describes how the fixed archetype behaves under selected weather stress, not how a probabilistic controller would perform.

The weather panel contains 144 annual weather files: six cities (Ahmedabad, Beijing, Guangzhou, Houston, Kolkata, and Phoenix), two emissions scenarios (SSP2-4.5 and SSP5-8.5), four time slices (baseline 2020s, near 2030s, mid 2050s, and late 2080s), and three selected annual severities (typical, hot, and heatwave-extreme). Each annual run produces 35,040 15-minute timesteps, of which 16,704 are occupied under the weekday office schedule. Across the full panel, the diagnostic therefore contains 2,405,376 occupied probability timesteps.

The future-weather files are used as stress-test inputs. They are not interpreted as a probabilistic forecast of any city-year, and the fixed Medium Office prototype is not treated as a forecast of future building stock. The experiment asks how a fixed building and fixed comfort-risk mapping behave under selected future-weather forcing from a warming-climate panel.

3.4. Zone aggregation

The Medium Office model contains 15 conditioned thermal zones. For each occupied timestep and zone, the trace recorder stores air temperature, mean radiant temperature, relative humidity, expected TSV, warm-tail probability, cold-tail probability, total discomfort-tail probability, and directional tail dominance. The main building-level aggregate is the simple mean across zones:

$$\bar{p}_{\text{tail}}^{(t)} = \frac{1}{|\mathcal{Z}|} \sum_{z \in \mathcal{Z}} p_{\text{tail},z}^{(t)}. \tag{9}$$

The zone audit compares this mean-zone value with the 90th-percentile zone value and the maximum zone value. Any-zone high-tail exceedance is defined at each occupied timestep as

$$I_{\text{any}}^{(t)} = \mathbb{1}\left\{\max_{z\in\mathcal{Z}} p_{\text{tail},z}^{(t)} \geq 0.20\right\}. \tag{10}$$

Hidden any-zone exceedance is the subset of occupied timesteps for which the mean-zone value is below the high-tail screen but at least one zone crosses it:

$$I_{\text{hidden}}^{(t)} = \mathbb{1}\left\{\bar{p}_{\text{tail}}^{(t)} < 0.20 \text{ and } \max_{z\in\mathcal{Z}} p_{\text{tail},z}^{(t)} \geq 0.20\right\}. \tag{11}$$

The maximum-zone value is used only as a diagnostic for modeled any-zone exceedance. It is not proposed as a universal building-control rule.

EnergyPlus zones are well-mixed air volumes. The zone-level diagnostic therefore does not resolve radiant asymmetry near glazing, vertical stratification, local air movement, or occupant-location-specific exposure. Any-zone exceedance is a modeled occupied-timestep zone-state metric, not a measured occupant-exposure metric. Its purpose is narrower: to quantify information loss when zone-level probability states are averaged.

3.5. Scalar PMV and expected-TSV comparison

The trace recorder also stores mean PMV across zones. We compare the primary p_{tail} output against two scalar summaries: $|\mu_{\text{TSV}}|$ and $|\text{PMV}|$. In that comparison, both scalars are evaluated against the same probability distribution produced by the primary TSV model. For each scalar, we report the standard threshold 0.5 and the best global scalar threshold for predicting $p_{\text{tail}} \geq 0.20$ in the 144-case panel. The goal is not to argue that PMV is invalid. PMV is used as a familiar scalar mean-state comparator, while μ_{TSV} is the scalar expectation from the same probability distribution that defines p_{tail}.

Because PMV is an engineered covariate in the primary TSV predictor, we also run a no-PMV robustness model. This is a separate ordinal TSV predictor trained with the same pipeline after removing PMV from the feature set, then re-applied to the same 144 simulated traces. In the robustness comparison, the x-axis PMV scalar remains the same stored thermal-state value, but the y-axis changes from the primary p_{tail} output to the no-PMV p_{tail} output. This design tests whether the observed scalar-reference pattern is created only by including PMV as a predictor feature.

We also report held-out TSV validation metrics for the primary predictor, the no-PMV predictor, and a deterministic PMV class baseline. The PMV baseline is computed by rounding PMV to the nearest seven-point TSV class and clipping the result to $[-3, +3]$. It is therefore a scalar-label baseline, not a calibrated probability model; log loss and expected-probability metrics are reported only for the probabilistic TSV predictors.

3.6. Metabolic-spread diagnostic

The base diagnostic uses fixed office defaults for occupant variables. To test what a single representative metabolic assumption hides, we perform an offline metabolic-spread diagnostic on the same simulated zone environments. The EnergyPlus thermal states are not rerun with altered internal heat gains. Instead, the TSV probability model is re-evaluated for each occupied zone state under six metabolic-load profiles derived from the published correction of the 120-W assumption [23]: 90.0, 93.2, 100.0, 108.0, 110.0, and 120.0 W/person. Under the paper-convention conversion, 120 W/person equals 1.2 met. These profiles are sensitivity cases for the reported-TSV model, not demographic strata assigned to the simulated occupants.

This diagnostic isolates representation sensitivity. It estimates how p_{tail} changes when the same simulated environment is interpreted under different plausible metabolic-load assumptions. It should not be read as a building-physics result for altered occupant heat gains.

3.7. Run quality assurance

The full zone-raw diagnostic run produced 144 per-case trace files, 144 EnergyPlus error files, and 144 EnergyPlus end files. Each trace contains 35,040 annual rows, 16,704 occupied rows, and 45 raw zone environmental fields for the 15 conditioned zones. No trace had missing or nonfinite occupied raw zone values. EnergyPlus completed all annual cases without fatal errors. Sixteen cases reported warmup load-convergence Severe messages; these cases still completed successfully and are retained with this QA caveat.

4. Results

4.1. Run integrity and panel scale

The full diagnostic-reference panel completed 144 annual EnergyPlus cases. Each trace contains 35,040 records and 16,704 occupied probability records, yielding 2,405,376 occupied records. The trace schema was consistent across cases, all 45 raw zone environmental fields were populated during occupied periods, and no fatal EnergyPlus errors occurred. Sixteen cases reported warmup-convergence Severe messages, but EnergyPlus completed successfully in each case. These messages are treated as simulation QA limitations rather than failed cases.

Because the diagnostic-reference strategy did not use the probability outputs to adjust setpoints, the results below describe representation behavior under fixed simulated environments. They should not be read as closed-loop HVAC outcomes.

4.2. Mean TSV and discomfort-tail probability

Across the full panel, mean p_{tail} was 0.117 and the 95th percentile was 0.306. Using the screening cutoff $p_{\text{tail}} \geq 0.20$, 13.4% of occupied records were high-tail states. High-tail exceedance increased with later time slices and higher scenario forcing (Table 2). The baseline-2020s slice had 9.17% high-tail exceedance, while the late-2080s slice had 21.18%. SSP2-4.5 had 11.75% high-tail exceedance, while SSP5-8.5 had 15.15%.

City differences were large. Kolkata had the highest mean-zone high-tail exceedance (19.90%), followed by Ahmedabad (17.11%), Phoenix (13.36%), Guangzhou (12.51%), Houston (12.33%), and Beijing (5.48%). The highest case-level exceedance rates occurred in

Table 2: High-tail exceedance summary for the 144-case fixed-reference panel. High-tail exceedance is the share of occupied records with $p_{\text{tail}} \geq 0.20$.

Grouping	Level	Mean p_{tail}	High-tail exceedance (%)
All	Full panel	0.117	13.45
Scenario	SSP2-4.5	0.112	11.75
Scenario	SSP5-8.5	0.121	15.15
Time slice	baseline_2020s	0.105	9.17
Time slice	near_2030s	0.105	9.16
Time slice	mid_2050s	0.118	14.28
Time slice	late_2080s	0.137	21.18
Severity	typical	0.109	10.48
Severity	hot	0.120	14.79
Severity	heatwave_extreme	0.121	15.07

hot or heatwave-extreme late-century files, including Kolkata SSP5-8.5 late-2080s hot and heatwave-extreme cases, both at 41.13% high-tail exceedance.

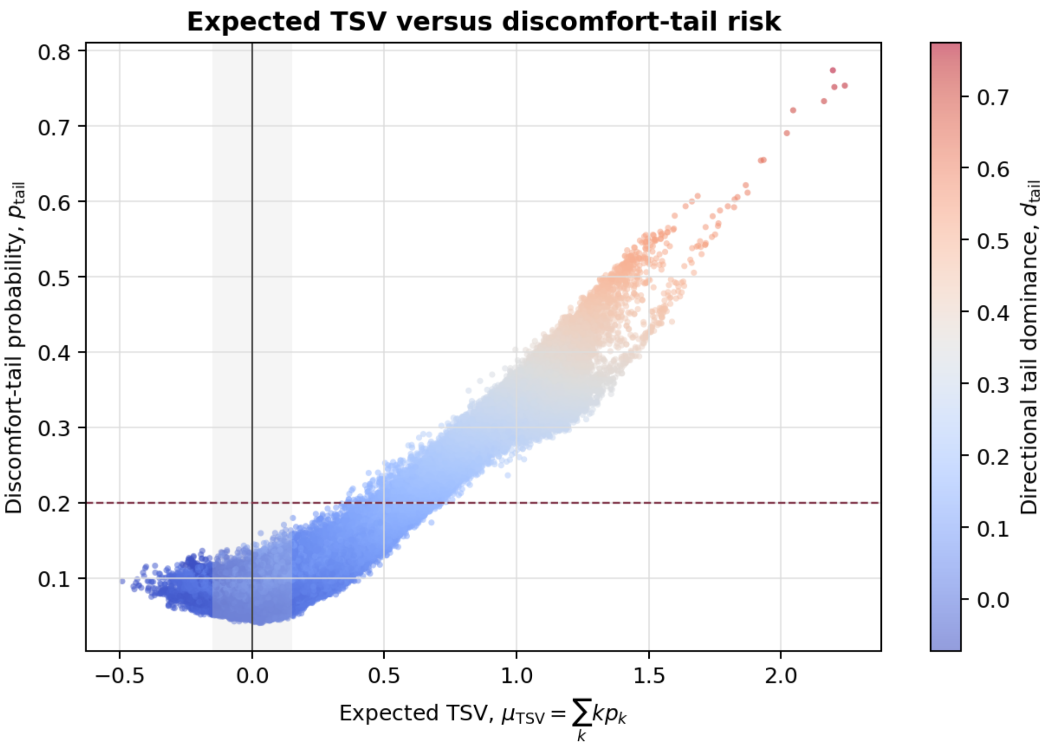

Figure 1: Expected TSV versus discomfort-tail probability for a sampled subset of the 144-case diagnostic panel. Expected TSV denotes the probability-weighted mean $\mu_{\text{TSV}} = \sum_k kp_k$, not the most likely TSV class. The horizontal line marks the screening cutoff $p_{\text{tail}} = 0.20$. Expected TSV is informative, but p_{tail} is the quantity that directly encodes the selected high-tail flag.

Figure 1 shows the internal relationship between the TSV probability distribution and two summaries derived from it. Expected TSV remained strongly associated with discomfort-tail probability, so the scalar mean is not meaningless. The diagnostic does not show near-neutral high-tail states under the strict condition $|\mu_{\mathrm{TSV}}| < 0.15$; that share was 0.0%. The stronger finding is different: similar positive mean states can still carry materially different tail probabilities. In the same-mean contrast examples, pairs with nearly equal μ_{TSV} differed in p_{tail} by up to about 0.18. For example, late-century Kolkata and Phoenix records with $\mu_{\mathrm{TSV}} \approx 1.35$ had p_{tail} values of 0.340 and 0.518, respectively. A scalar mean gives the warm-side tendency, but the probability distribution shows how much mass has accumulated in the severe tail.

The same figure also shows that tail risk in this fixed-reference panel is warm-dominant. Mean warm-tail probability was 0.089, compared with 0.027 for the cold tail. Warm-tail probability exceeded cold-tail probability in 78.9% of occupied records and in all records crossing $p_{\mathrm{tail}} \geq 0.20$. Records with $\mu_{\mathrm{TSV}} \leq 0$ never crossed the high-tail screen, while all records with $\mu_{\mathrm{TSV}} > 1$ did. Thus, above about +1 expected TSV, the scalar mean already identifies a warm-risk state; the probability distribution is still needed to quantify how much severe-tail mass is present.

4.3. PMV as a scalar benchmark

The PMV comparison gives a second scalar benchmark. Across the same occupied records, |PMV| was correlated with p_{tail} (Pearson $r = 0.856$, Spearman $\rho = 0.751$), while $|\mu_{\mathrm{TSV}}|$ was more tightly correlated with p_{tail} (Pearson $r = 0.976$, Spearman $\rho = 0.896$). A best offline $|\mu_{\mathrm{TSV}}|$ threshold of 0.609 disagreed with the high-tail flag in 1.09% of occupied records. A best offline |PMV| threshold of 0.912 disagreed in 3.37% of occupied records.

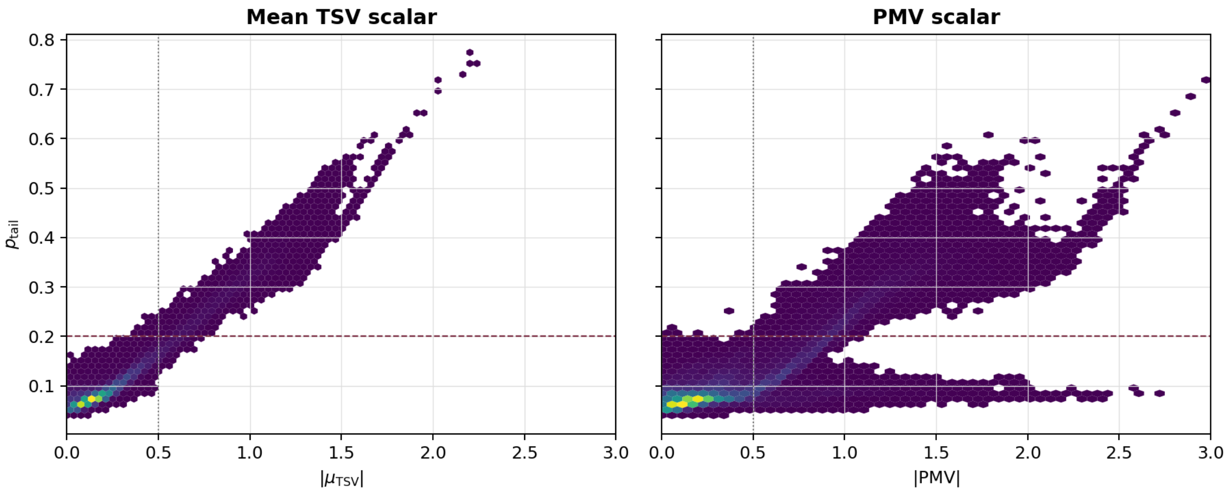

Figure 2: Scalar comparison against the primary TSV probability output. Left: $|\mu_{\mathrm{TSV}}|$ versus p_{tail}. Right: |PMV| versus the same p_{tail} distribution. Both panels use the same 0–3 scalar range. The vertical line marks the conventional 0.5 scalar band, and the horizontal line marks the diagnostic high-tail screen $p_{\mathrm{tail}} = 0.20$.

Table 3: Scalar threshold comparison against the $p_{\text{tail}} \geq 0.20$ high-tail flag. Best thresholds are optimized offline over the full diagnostic panel. Missed high-tail and false-positive rates are reported as percentages of all occupied records.

Scalar rule	**Threshold**	**Disagreement (%)**	**Missed high-tail (% records)**	**False positive (% records)**
$\lvert\mu_{\text{TSV}}\rvert$ best threshold	0.609	1.09	0.62	0.47
$\lvert\text{PMV}\rvert$ best threshold	0.912	3.37	1.56	1.81
$\lvert\text{PMV}\rvert$ standard band	0.500	19.11	0.006	19.10

Read together, Figures 1 and 2 separate two issues. Figure 1 shows how the learned TSV expectation relates to its own tail probability. Figure 2 asks how that relationship differs from PMV, Fanger's continuous comfort index built from thermal-balance terms and empirical chamber-vote evidence. The standard $|\text{PMV}| = 0.5$ band behaved conservatively in this panel. It almost never missed high-tail states, but it flagged many states that did not cross $p_{\text{tail}} \geq 0.20$. The learned expected-TSV scalar occupies a narrower range and follows the calibrated tail probability closely because both are summaries of the same seven-class probability vector. This is a tighter empirical association, not a claim of linearity: $|\mu_{\text{TSV}}|$ reports the location of the predicted distribution, while p_{tail} reports its severe-category mass. Its maximum absolute value was 2.24 in this panel, and only 3.7% of occupied records exceeded $|\mu_{\text{TSV}}| = 1$. PMV spans a wider scalar range, reaching 3.31 in absolute value, and includes a lower-p_{tail} branch at higher $|\text{PMV}|$ values. This broad branch around $p_{\text{tail}} \approx 0.1$ is the visual source of the conservative PMV result: PMV can assign a large scalar departure to states that the calibrated TSV distribution still places mostly in neutral or mild categories. Among records with $|\text{PMV}| \geq 0.5$, 58.7% remained below $p_{\text{tail}} = 0.20$; even among records with $|\text{PMV}| \geq 1$, 9.6% remained below that tail-probability screen. The mid-tail band shows the same compression: for records with $0.3 \leq p_{\text{tail}} < 0.4$, the 5th–95th percentile range was 0.88–1.20 for $|\mu_{\text{TSV}}|$ but 1.04–1.86 for $|\text{PMV}|$. The point is not that PMV is useless or blind to thermal stress; rather, PMV and p_{tail} answer different diagnostic questions. PMV gives a conventional continuous mean-vote index for the thermal state. p_{tail} reports the predicted probability mass in severe TSV categories.

The no-PMV robustness check gave the same qualitative result, but it should be read as a model-output comparison rather than as two versions of PMV. Figure 3 keeps the x-axis fixed as the same stored $|\text{PMV}|$ scalar and changes the y-axis from the primary p_{tail} output to the p_{tail} output from a separately trained no-PMV TSV predictor. Removing PMV from the ordinal predictor changed the mean-zone high-tail exceedance at $p_{\text{tail}} \geq 0.20$ from 13.45% to 13.85%, and changed any-zone exceedance from 37.03% to 37.52%. The PMV-inclusive output is modestly tighter against the PMV scalar, with Pearson $r = 0.856$ and Spearman $\rho = 0.751$, compared with $r = 0.849$ and $\rho = 0.726$ for the no-PMV output. This supports a limited interpretation: calculable PMV is a useful predictor feature for reported TSV probabilities, but the main scalar-reference and aggregation results are not created solely by including PMV in the model. Under the no-PMV predictor, $|\mu_{\text{TSV}}|$ still had Pearson $r = 0.976$ with its own p_{tail} output. We therefore treat the PMV comparison as a descriptive scalar-reference audit, not as the main evidence for the probability diagnostic.

The held-out TSV validation check gives the same practical reading (Tables 4 and 5). The

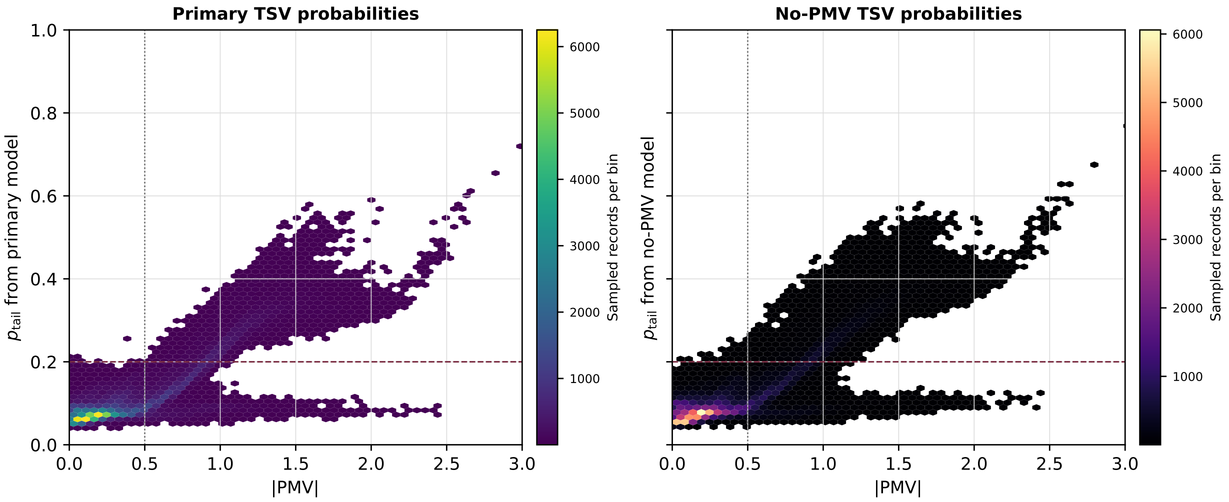

Figure 3: PMV-scalar robustness check using two TSV probability outputs. Both panels use the same x-axis scalar, |PMV|, from the simulated thermal state. The left panel plots PMV against p_{tail} from the primary TSV predictor, which includes PMV as an engineered covariate. The right panel plots the same PMV values against p_{tail} from a separately trained no-PMV TSV predictor. The different palettes indicate that the two panels contain different probability-model outputs, not different PMV calculations.

primary TSV predictor was slightly stronger than the no-PMV predictor, but the difference was small: exact-class accuracy changed from 47.3% to 46.9%, within-one-class accuracy from 84.1% to 83.8%, and ordinal class MAE from 0.73 to 0.73. The rounded-PMV class baseline was weaker on exact and ordinal metrics, with 33.2% exact accuracy and 0.97 class MAE. Class-specific recall was modest in the tail classes for all methods, which is consistent with the neutral-heavy and noisy structure of field TSV data. Supplementary calibration checks show that aggregate predicted p_{tail} aligned closely with observed tail frequency on the held-out split: the primary model's mean predicted p_{tail} was 0.207 versus an observed tail frequency of 0.206, with fixed-bin tail ECE of 0.011. We also tested a PMV-only calibrated tail baseline, fitted as $|\text{PMV}| \to \Pr(\text{TSV} \leq -2 \text{ or } \text{TSV} \geq +2)$ on the same non-test data. Table 4 shows that this baseline matched overall tail prevalence on the held-out split, but it discriminated tail records less well than the TSV probability models. AUROC was 0.571 versus 0.747, average precision was 0.260 versus 0.483–0.487, Brier score was 0.161 versus 0.137–0.138, and F1 at the $p_{\text{tail}} \geq 0.20$ screen was 32.6% versus 46.4–47.1%. A stricter contributor-grouped robustness check in the supplement showed weaker transfer, especially for tail-class discrimination, which limits any absolute reading of simulated tail-risk percentages under domain shift. It also showed only small differences between the primary and no-PMV feature sets. The validation evidence therefore supports a limited claim: PMV is useful enough to improve the primary ordinal predictor slightly, and PMV alone can be calibrated to the overall tail prevalence, but the 144-case diagnostic patterns should not be treated as a PMV-only artifact.

Table 4: Held-out tail-probability comparison for the TSV probability models and the PMV-only calibrated scalar baseline. F1 is computed by screening each predicted tail probability at $p_{\text{tail}} \geq 0.20$.

Predictor	**Mean predicted**	**Observed**	**Brier**	**ECE**	**F1 (%)**	**AUROC**
Primary TSV model	0.207	0.206	0.137	0.011	47.1	0.747
No-PMV TSV model	0.207	0.206	0.138	0.006	46.4	0.747
PMV-only calibrated tail baseline	0.206	0.206	0.161	0.003	32.6	0.571

Table 5: Held-out TSV class validation for the primary predictor, the no-PMV robustness predictor, and a rounded-PMV class baseline. The PMV baseline rounds PMV to the nearest TSV class and clips to $[-3, +3]$; it is a deterministic scalar-label baseline, not a probability model. Class recall columns report the share of records in each observed TSV class assigned to that same class.

Predictor	**Exact (%)**	**Within ±1 (%)**	**Class MAE**	**Tail F1 (%)**	**Log loss**	**TSV -3**	**TSV -2**	**TSV -1**	**TSV 0**	**TSV +1**	**TSV +2**	**TSV +3**
Primary TSV model	47.3	84.1	0.73	34.0	1.454	25.5	6.9	12.4	88.0	16.2	15.7	23.9
No-PMV TSV model	46.9	83.8	0.73	33.5	1.476	23.7	6.5	13.0	87.4	15.4	15.8	23.2
Rounded PMV class	33.2	76.5	0.97	26.6	–	13.8	10.4	27.6	47.8	26.3	14.5	10.5

4.4. Zone aggregation hides any-zone tail exceedance

Mean-zone aggregation produced a stable building-level summary, but it also hid zone-level risk. Across the full panel, mean-zone aggregation identified 13.45% high-tail exceedance. Any-zone aggregation identified 37.03%, and p90-zone aggregation identified 28.19%. These are modeled occupied-timestep zone-state exceedances, not measured occupant exposures. In 23.58% of occupied records, the mean-zone value was below $p_{\text{tail}} = 0.20$ while at least one zone exceeded it. A maximum-zone summary would assign a more severe tail category than the mean in 31.69% of occupied records; a p90-zone summary would do so in 20.94%.

Figure 4 tests whether this hidden-zone effect depends on the selected reporting screen. It repeats the exceedance calculation from $\tau = 0.05$ to $\tau = 0.40$, where τ is the high-tail screen applied to p_{tail}. Absolute exceedance falls as the screen becomes stricter, but any-zone exceedance remains well above mean-zone exceedance across the range. For example, at $\tau = 0.10$, mean-zone exceedance was 35.74% and any-zone exceedance was 60.78%; at $\tau = 0.30$, they were 5.46% and 24.10%, respectively. Hidden any-zone exceedance remained between 18.63% and 25.32% for thresholds from 0.10 to 0.30. The no-PMV predictor showed the same pattern: at $\tau = 0.20$, mean-zone exceedance was 13.85%, any-zone exceedance was 37.52%, and hidden any-zone exceedance was 23.67%.

The area-weighted zone-time diagnostic gives a useful check on this result. When each zone-time record was weighted by conditioned floor area, high-tail exceedance was 13.17%, close to the mean-zone value. This does not remove the local risk signal. It shows that much of the hidden any-zone exceedance comes from smaller perimeter zones rather than from the largest core zones.

The highest-risk zones were perimeter zones, especially P1, the south perimeter orientation in this DOE Medium Office geometry. The largest high-tail exceedance rates were 30.51% for `perimeter_bot_zn_1`, 27.91% for `perimeter_mid_zn_1`, and 23.59% for `perimeter_top_zn_1`. Each of these zones accounts for about 4.16% of conditioned floor area, while each core zone accounts for about 19.74%. Thus, the most severe modeled local

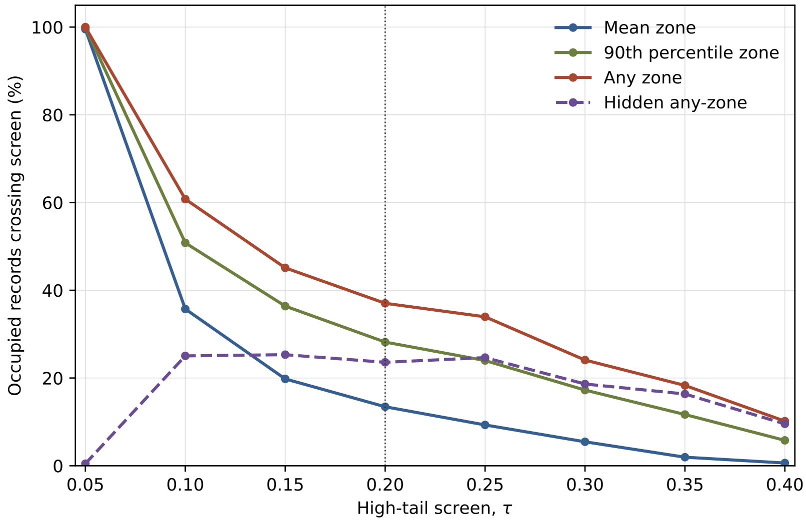

Figure 4: Threshold sensitivity of the hidden-zone aggregation effect across the full 144-case fixed-reference panel: six cities, SSP2-4.5 and SSP5-8.5, four time slices, and three annual weather severities. The figure is pooled across pathways and is not city- or pathway-specific. The vertical line marks the manuscript reporting screen $\tau = 0.20$. The persistent gap between mean-zone, p90-zone, and any-zone exceedance shows that the zone-aggregation result is not specific to a single threshold.

Table 6: Zone aggregation summary by city. Mean, any-zone, and p90-zone columns report modeled high-tail exceedance at $p_{\text{tail}} \geq 0.20$. Any-zone exceedance is the share of occupied timesteps for which the maximum zone value satisfies $\max_z p_{\text{tail},z} \geq 0.20$. Hidden high-tail exceedance means the mean-zone value is below 0.20 while at least one zone is at or above 0.20.

City	Mean-zone (%)	Area-weighted zone-time (%)	Any-zone (%)	p90-zone (%)	Hidden any-zone (%)
Ahmedabad	17.11	17.66	54.48	42.16	37.37
Beijing	5.48	5.86	15.55	9.98	10.07
Guangzhou	12.51	11.92	32.08	23.27	19.56
Houston	12.33	11.06	28.18	22.31	15.85
Kolkata	19.90	20.73	47.27	36.36	27.36
Phoenix	13.36	11.81	44.64	35.07	31.28

tail exceedance occurs in relatively small perimeter zones, while the building-level area-weighted signal is shaped strongly by the larger core zones.

Figure 5 shows that this local pattern changes by city and time slice. Beijing remains comparatively low-risk across the panel: its maximum zone-mean high-tail exceedance rises from 8.8% in the baseline-2020s slice to 18.6% in the late-2080s slice. Guangzhou shows a clearer future-weather transition, with the maximum zone value increasing from 23.1% in the baseline-2020s slice to 34.3% in the mid-2050s slice and 53.1% in the late-2080s slice. Kolkata reaches high exceedance earlier: its maximum zone value is already 39.2% in the baseline-2020s slice and reaches 65.0% in the mid-2050s slice. Ahmedabad has a different structure: P1 is already highly exposed, but non-P1 perimeter and core zones remain much lower until the late-2080s slice. This is why Ahmedabad can look severe under an any-zone view while the broader zone field is less uniformly severe than Kolkata.

Figure 6 was generated as a direct check on the floor/orientation reading from Fig. 5. It prevents over-reading the mosaic as a simple floor-height effect. Area-weighted high-tail exceedance was 11.04% for the bottom floor, 15.11% for the middle floor, and 13.36% for the top floor. The middle floor had the highest area-weighted floor exceedance in 135 of 144 cases. Bottom-floor exceedance was highest only under an any-zone-within-floor summary, where the south perimeter zone P1 dominated. Bottom P1 had 30.51% high-tail exceedance, compared with 27.91% for middle P1 and 23.59% for top P1. For the other perimeter orientations, bottom was not the highest. We therefore interpret the floor pattern as a prototype-specific interaction between perimeter orientation, floor construction, and zone conditioning, not as a general result that lower floors are more exposed than upper floors. High-tail states were warm-dominant in all zones under the $p_{\text{tail}} \geq 0.20$ screen.

4.5. Metabolic-spread sensitivity

The metabolic-spread diagnostic shows that representative occupant assumptions can materially change the tail-risk profile even when the simulated thermal environments are held fixed. Table 7 is a panel-level summary across both SSP pathways, all six cities, four time slices, and three annual weather severities; it is not a pathway-specific estimate. Across 216,483,840 zone-profile probability evaluations, the mean spread in mean-zone p_{tail} across the six metabolic profiles was 0.074, with a 95th percentile spread of 0.209. For max-zone p_{tail}, the mean spread was 0.137 and the 95th percentile spread was 0.327.

Changing the metabolic profile also changed threshold status. Mean-zone high-tail status flipped across metabolic profiles in 18.99% of occupied records, and max-zone high-tail status flipped in 38.66%. When the S-real profile remained below the high-tail cutoff, some other metabolic profile crossed it in 18.80% of mean-zone records and 36.51% of any-zone records. Because the fixed-reference panel is warm-tail dominated, lower metabolic-load profiles generally reduced predicted high-tail exposure. In this diagnostic, a lower metabolic input means the same simulated thermal environment is interpreted by the reported-TSV model with less occupant metabolic heat.

The response is not strictly monotone across all adjacent profiles; in particular, the calibrated model gives lower mean-zone high-tail exposure at 110 W/person than at 108 W/person. We therefore interpret this section as a sensitivity diagnostic across metabolic

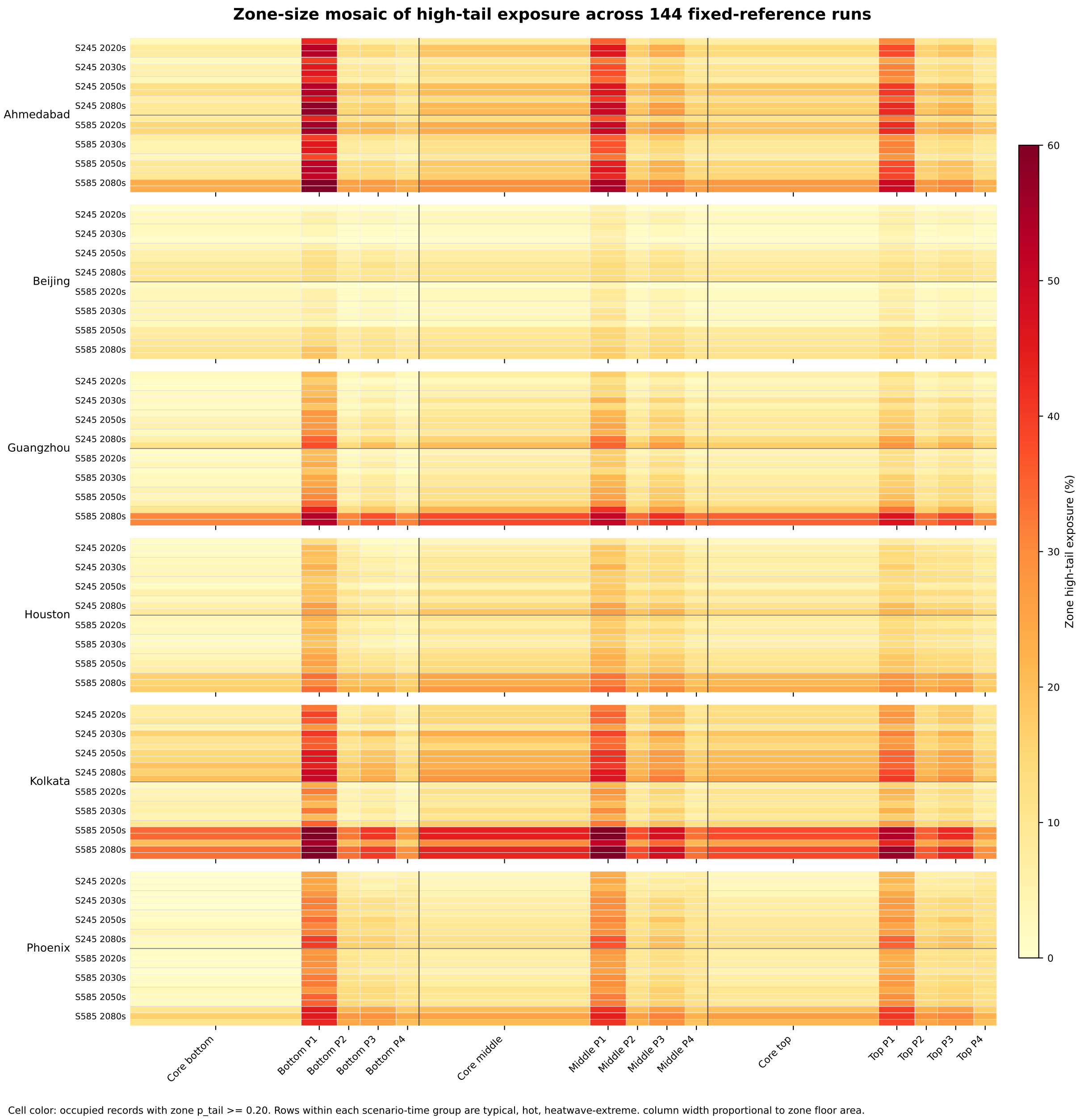

Figure 5: Zone-size mosaic of modeled high-tail exceedance across the 144-case fixed-reference panel. Cell color reports the share of occupied records in each case-zone pair with $p_{\text{tail}} \geq 0.20$. Column width is proportional to conditioned floor area, so the figure compares local tail exceedance with the physical size of the zone carrying that signal. Core bottom, core middle, and core top are the three core zones of the DOE Medium Office prototype; Bottom P1–P4, Middle P1–P4, and Top P1–P4 are the four perimeter zones on each floor level. P1–P4 denote perimeter orientations: P1 south, P2 east, P3 north, and P4 west. The conditioned top zones sit below the top-floor plenum and roof assembly, so "top" should not be read as a directly roof-exposed room condition. Rows within each scenario-time group are ordered as typical, hot, and heatwave-extreme weather files.

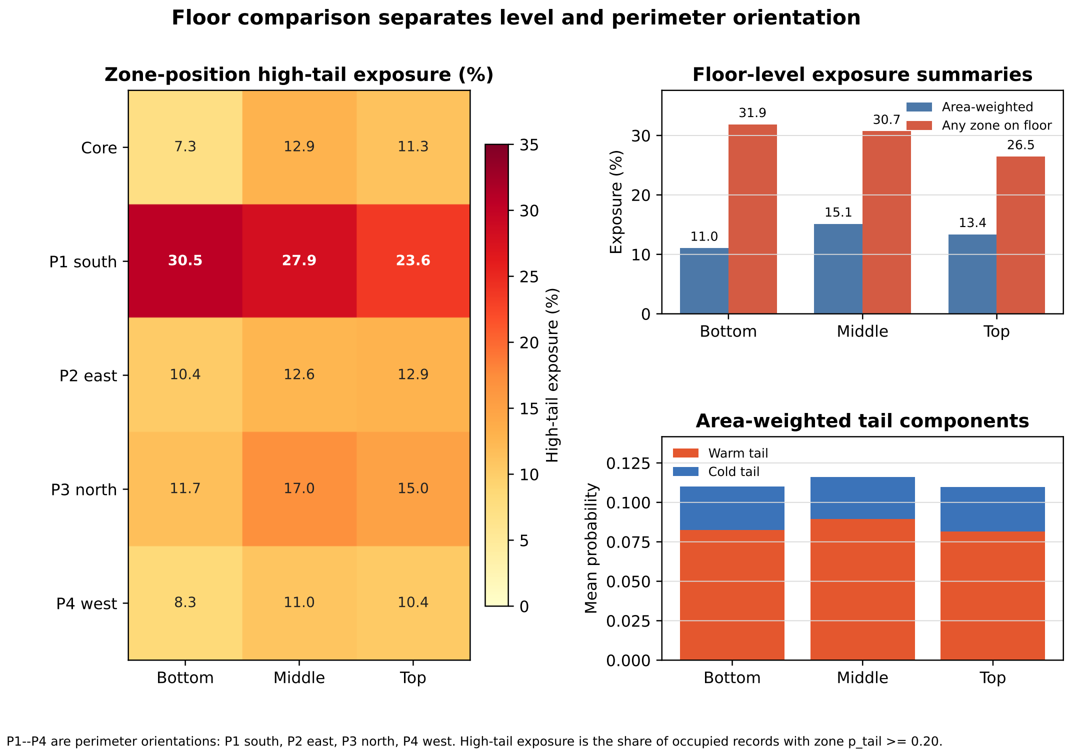

Figure 6: Floor-position comparison for the 144-case fixed-reference panel. The heat map reports modeled high-tail exceedance by floor and zone position. P1–P4 are the perimeter orientations in the DOE Medium Office prototype: P1 south, P2 east, P3 north, and P4 west. Area-weighted floor exceedance summarizes zone-time records using conditioned floor area, while any-zone floor exceedance reports the share of occupied records in which at least one zone on that floor exceeds $p_{\text{tail}} \geq 0.20$. Tail-component bars show that floor differences are driven mainly by the warm tail.

Table 7: Panel-level metabolic-spread results across the full 144-case fixed-reference panel: six cities, SSP2-4.5 and SSP5-8.5, four time slices, and three annual weather severities. Values are pooled across pathways and recomputed over fixed environmental traces; EnergyPlus was not rerun with altered occupant heat gains.

Profile	**W/person**	**met**	**Mean-zone high-tail (%)**	**Any-zone high-tail (%)**
COMP-Lo	90.0	0.900	3.32	13.82
S-real	93.2	0.932	3.10	14.89
COMP-Med	100.0	1.000	7.97	29.78
EQ-Max	108.0	1.080	21.20	45.37
COMP-Hi	110.0	1.100	13.66	38.44
LEGACY	120.0	1.200	17.32	47.87

assumptions, not as a smooth physiological dose-response curve. The practical result is still clear: the single representative occupant assumption can hide a substantial range of predicted reported-TSV tail exposure.

The zone-resolved profile summaries show where this sensitivity is concentrated. Across the 2,160 case-zone summaries for each profile, mean case-zone high-tail exposure was 3.27% for COMP-Lo, 3.26% for S-real, 8.65% for COMP-Med, 19.25% for EQ-Max, 15.37% for COMP-Hi, and 18.77% for LEGACY. The spread in case-zone high-tail exposure across the six profiles averaged 16.85 percentage points, with a 95th percentile of 36.15 percentage points. The largest mean spreads occurred in the south perimeter zones: Bottom P1 (30.44 percentage points), Middle P1 (28.64 percentage points), and Top P1 (24.96 percentage points). This reinforces the diagnostic claim: metabolic assumptions and zone aggregation interact, so a single representative occupant can hide spatially localized profile sensitivity.

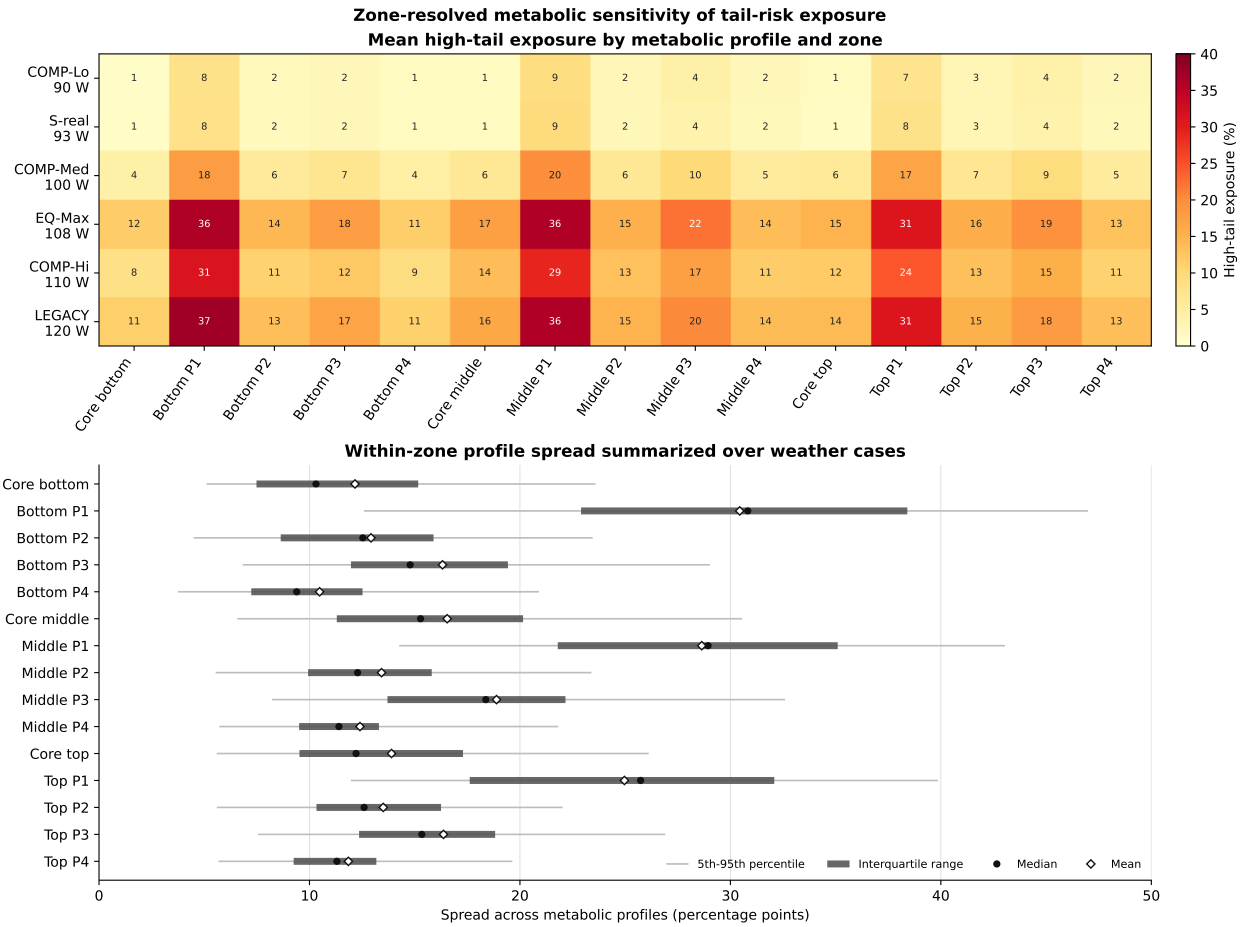

Figure 7: Zone-resolved metabolic sensitivity of high-tail exposure. Top: mean case-zone high-tail exposure by metabolic profile and zone; color encodes high-tail exposure only. Bottom: within-zone spread across metabolic profiles summarized over the 144 weather cases. Points mark medians, diamonds mark means, thick intervals show the interquartile range, and thin intervals show the 5th–95th percentile range. Values are recomputed over fixed simulated environments, so the differences reflect representation sensitivity to metabolic inputs rather than altered internal heat gains.

5. Discussion

5.1. What the probability diagnostic adds

The main contribution is not a new HVAC method. It is a diagnostic interface for reading a calibrated TSV probability distribution. Expected TSV, defined as the probability-weighted mean $\mu_{\mathrm{TSV}} = \sum_k kp_k$, remains useful: in the full panel it was strongly associated with p_{tail}, and the diagnostic did not produce near-neutral high-tail states under $|\mu_{\mathrm{TSV}}| < 0.15$. This is an important constraint on the interpretation. The probability representation does not overturn the scalar mean.

The added value is that the distribution makes tail probability explicit. A scalar mean reports central tendency; p_{tail} reports the probability mass in the severe cold and warm categories. Similar mean states can still have different tail probabilities, and the threshold status of p_{tail} is directly available only when the distribution is retained. In this fixed-reference future-weather panel, the visible tail was also strongly warm-skewed: cold-tail probability was small, and high-tail records were warm-dominant. That pattern is conditional on the selected cities, weather files, fixed thermostat schedule, and static comfort model, but it is exactly the kind of directional diagnostic that the probability object makes explicit.

The panel summary in Table 2 also matters because it separates broad stress-test gradients from local case differences. High-tail exceedance increased from the baseline-2020s slice to the late-2080s slice and was higher under SSP5-8.5 than SSP2-4.5, but the hot and heatwave-extreme annual files were close in aggregate exceedance. The city spread was larger than either scenario contrast or annual-severity contrast. This supports a cautious reading: the fixed archetype shows more high-tail states under later and higher-forcing weather, but the diagnostic should not be reduced to a single climate-scenario ranking. Local climate, selected annual file, and building-zone response still shape the observed tail profile.

5.2. PMV is a comparator, not a strawman

The PMV results also support a restrained interpretation. PMV was correlated with p_{tail}, and the standard $|\mathrm{PMV}| = 0.5$ band was conservative in this fixed-reference panel. It almost never missed high-tail states. Its main difference from p_{tail} was over-flagging: many states outside the PMV neutral band did not cross the selected TSV tail-probability cutoff. The comparison is also geometrically informative. The learned expected-TSV scalar stays within a narrower empirical range and tracks the calibrated tail probability closely, while PMV spans a wider continuous comfort-index range and includes a lower-tail-probability branch at elevated $|\mathrm{PMV}|$. This tighter mapping is unsurprising because μ_{TSV} and p_{tail} are two summaries of the same calibrated TSV probability vector; it should not be interpreted as a claim that the relationship is linear. That branch means that some states can look substantially displaced on the PMV scale while the calibrated TSV distribution still assigns most probability to neutral or mild bins. This is expected: μ_{TSV} is a probability-weighted mean of the calibrated seven-class TSV predictor used in this study, whereas PMV is Fanger's continuous index built from thermal-balance terms and empirical chamber-vote evidence. The two scalars therefore stretch the same simulated thermal states differently.

Table 3 makes the threshold consequence explicit. An offline optimized |PMV| threshold can be made closer to the p_{tail} screen, but it still disagrees more often than the optimized expected-TSV threshold. The conventional |PMV| = 0.5 band has a different purpose: it is protective as a comfort band, not tuned to reproduce this paper's tail-probability screen. The high false-positive rate under that band is therefore not a defect by itself; it shows that a conventional scalar comfort band and a calibrated severe-tail probability flag are not interchangeable diagnostics.

This means the diagnostic should not be framed as PMV failure. PMV remains a valuable standard comfort index and benchmark. The primary TSV model also uses PMV as one covariate, so the PMV panel is not an independent model contest. The no-PMV robustness check is therefore important, but it has a narrow meaning: the same stored PMV scalar was compared against two different TSV probability outputs, one from the primary model and one from a separately trained model without PMV. Removing PMV modestly loosened the PMV–p_{tail} relationship, which is consistent with PMV carrying useful information for reported TSV prediction, but it left the main aggregation result nearly unchanged. The probability diagnostic answers a different question: how much of the calibrated empirical TSV distribution lies in discomfort tails? For building studies that need a compact scalar, PMV or expected TSV may be enough. For studies that need tail-risk screening, the probability vector carries information that scalar summaries do not directly expose.

5.3. Zone aggregation is a major source of information loss

The zone results show that aggregation can matter more than the choice between two scalar summaries. Mean-zone aggregation gave 13.4% modeled high-tail exceedance, while any-zone aggregation gave 37.0%. This is not a minor reporting difference; it changes the apparent risk profile of the same fixed simulated building. The threshold sweep strengthens this point. Changing the high-tail screen changes the absolute exceedance percentage, as expected, but it does not remove the gap between mean-zone and any-zone summaries.

The zone pattern should not be summarized as a simple "top floor is worst" or "bottom floor is worst" result. In this prototype, P1 is the south perimeter zone and is the dominant local exceedance zone; bottom P1 was higher than middle and top P1 in most cases. At the same time, area-weighted floor exceedance was usually highest on the middle floor because the large core zone and the other perimeter orientations contributed more to the floor-level average. The any-zone floor metric answers a different question: whether at least one zone on that floor crosses the high-tail screen. It therefore exposes small-zone perimeter risk rather than suppressing it. This is why bottom-floor any-zone exceedance can be high even when the middle floor has the higher area-weighted mean. The conditioned top zones are also not direct roof-surface rooms; they sit below a top-floor plenum and roof assembly. The useful interpretation is therefore more specific: orientation, floor construction, zone size, aggregation rule, and local climate interact, and the probability diagnostic makes those interactions visible.

Figure 6 also shows that floor differences are differences in degree, not separate thermal regimes. The bottom, middle, and top floor summaries are broadly comparable in magnitude, with the middle floor slightly elevated under area weighting. The tail-component bars show the same directional structure on all three floors: warm-tail probability dominates the discomfort signal, while cold-tail probability remains small. This supports the paper's diagnostic framing. The main finding is not that one floor is categorically unsafe, but that the conclusion changes depending on whether the analyst reports a mean, area-weighted, percentile, or any-zone view.

The result should still be read carefully. EnergyPlus zone states are well-mixed summaries, so the diagnostic does not prove that occupants at specific desks experienced radiant asymmetry, stratification, or local drafts. Any-zone exceedance is therefore a modeled zone-state screen, not a measured occupant-exposure rate. It shows that even at the modeled zone level, averaging can hide perimeter/core differences that are relevant to a probability-state diagnostic. A future implementation should report mean, percentile, and max-zone summaries together, or keep the probability object at the zone level when the building system can act by zone.

5.4. Metabolic spread changes the risk profile

The metabolic-spread diagnostic turns a common modeling convention into a measurable uncertainty. Holding the environment fixed while changing the metabolic input produced threshold flips in 19.0% of mean-zone records and 38.7% of max-zone records. This does not mean that EnergyPlus loads would be unchanged in a real population; the diagnostic deliberately did not rerun internal heat gains. It isolates the representation question: how sensitive is the predicted TSV tail probability to the representative metabolic assumption?

The answer is that the sensitivity is large enough to affect the risk profile. This is especially relevant because metabolic rate is often treated as a fixed office default. In the warm-dominant fixed-reference panel, lower metabolic-load assumptions generally lowered predicted tail exceedance because the same environmental state is interpreted with less occupant heat production. That is the model-output result. The demographic interpretation is more cautious. The metabolic-rate correction used to define the profiles shows that lower W/person values may correspond to population groups such as older or smaller-bodied occupants [23]. The result should not be read as evidence that those occupants are safer or experience less actual discomfort. A recent review of elderly thermal characteristics emphasizes that older adults' reported thermal sensation and discomfort responses can differ from those of younger adults, including altered sensitivity and tolerance patterns [24]. The present diagnostic therefore estimates the probability of reporting severe TSV categories under a trained TSV model; it does not measure latent discomfort, physiological heat strain, or health vulnerability. The non-monotone response between some adjacent profiles also cautions against overinterpreting the model as a smooth physiological curve. Plausible metabolic assumptions expose a spread in predicted reported-TSV tail risk that a single mean-occupant convention hides.

5.5. Downstream use without overclaiming

The probability object could support future operational designs. A fuzzy rule base could use calibrated tail probabilities as inputs. A stochastic or chance-constrained formulation could use p_{tail} or cumulative TSV probabilities as constraints. A building operator could use the same diagnostic to screen weather files, zones, or occupant-assumption scenarios before choosing an operating policy.

Those are downstream uses, not results of this paper. The fixed-reference simulations here do not test energy savings, comfort improvement, setpoint stability, equipment wear, demand response, or field deployment. The 15-minute records are diagnostic probability records. They should not be interpreted as equipment commands.

6. Limitations and Future Work

Static comfort calibration. The TSV probability model is calibrated from contemporary ASHRAE Global Thermal Comfort Database II observations and is held fixed across all future-weather files. The study does not model future changes in expectations, clothing behavior, adaptive opportunity, physiology, culture, or thermal history. The results are therefore conditional stress-test diagnostics, not forecasts of how occupants will perceive buildings in the 2050s or 2080s.

Fixed building testbed. The DOE Medium Office prototype is held fixed across all cases. This isolates the weather and representation effects, but it does not represent future envelope upgrades, code changes, adaptive glazing, shading retrofits, HVAC redesign, or future operation practice.

Weather-panel scope. The 144 weather files are selected annual cases from a local CMIP-derived panel. They support comparisons across city, scenario, time slice, and severity, but they are not a probabilistic climate ensemble. The paper does not estimate event frequencies or regional climate uncertainty.

EnergyPlus zone resolution. EnergyPlus provides well-mixed zone-level states. The zone-aggregation diagnostic therefore quantifies information loss between modeled zones. It does not resolve sub-zone radiant asymmetry, vertical stratification, local air speed, solar patches, or occupant-location-specific exposure. Reported any-zone exceedance rates should therefore be read as modeled zone-state exceedances rather than as measured occupant-exposure rates.

Tail calibration. The discomfort tails are data-poor relative to the neutral class, especially at TSV = ± 3. Post-hoc isotonic calibration improves probability reliability, but tail estimates remain more uncertain than central categories. Future work should test tail-weighted training, partial proportional-odds models, mixture-of-threshold models, and calibration methods that better preserve class dependence in sparse tails.

Metabolic-spread interpretation. The metabolic diagnostic recomputes TSV probabilities over fixed environmental traces. It does not rerun EnergyPlus with altered occupant sensible heat gains, nor does it model correlations among metabolic rate, clothing, activity, age, sex, body size, or adaptation. The target is also reported TSV, not latent physiological strain or actual health risk. If some groups systematically report thermal sensation or discomfort differently from younger reference populations, or if TSV datasets encode reporting and adaptation biases, the diagnostic would inherit those biases. Lower predicted reported-TSV tail probability for a lower metabolic-rate profile should therefore not be equated with lower vulnerability. A full population model should represent these variables jointly rather than as independent scenario shocks and should distinguish reported sensation, comfort, preference, and physiological heat stress.

No closed-loop validation. The present study intentionally excludes closed-loop operating validation. The diagnostic does not establish better HVAC performance, lower EUI, improved comfort, setpoint feasibility, equipment stability, or reduced demand. Those claims would require separate closed-loop experiments with explicit objectives, equipment constraints, dwell-time logic, and field or high-fidelity validation.

7. Conclusions

This paper presents a diagnostic-interface study of probabilistic thermal comfort risk. It is not a control study. The central object is the calibrated seven-class TSV distribution, from which expected TSV and discomfort-tail probability are both derived. The key definition is simple: $p_{\text{tail}} = \Pr(\text{TSV} \leq -2) + \Pr(\text{TSV} \geq +2)$, the predicted probability mass in the cold and warm discomfort tails.

In a 144-case fixed-reference Medium Office panel, 13.4% of occupied records crossed $p_{\text{tail}} \geq 0.20$. High-tail exceedance increased in later time slices and under SSP5-8.5, showing how the fixed archetype appears under selected warming-climate weather stress. PMV and expected TSV both tracked tail risk, but neither replaces the probability distribution when the diagnostic question is explicitly about tail probability. Zone aggregation created the largest information loss: any-zone high-tail exceedance was 37.0%, compared with 13.4% under mean-zone aggregation. Metabolic-rate assumptions also changed threshold status, especially under max-zone aggregation.

The defensible conclusion is narrow and useful. Calibrated TSV probabilities reveal discomfort-tail, spatial, and occupant-assumption risk profiles that scalar comfort states compress. The result is a diagnostic representation that can inform future operational design, but it is not itself a validated operating strategy.

Data and Code Availability

The reproducibility workspace for this manuscript contains the run manifest, analysis scripts, processed diagnostic summaries, figure-generation inputs, model artifacts, weather-file provenance, trace-schema checks, and EnergyPlus run-quality audits used to generate the reported tables and figures. The public submission repository is `https://github.com/clementinec/probabilistic-comfort-risk-reproducibility`. It archives compact reproducibility materials and provides checksums for large traces and weather files where redistribution is not practical or not permitted. The ASHRAE Global Thermal Comfort Database II is a third-party dataset available from its original source subject to its terms of use and cannot be redistributed by the authors.

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